

Feature 233

energy + causal steering

Basic/acidic after an LSE/LSEE motif

Top activating sites	activation
LLSEEE[R]KEREKI	12.3
ELSEEE[K]EEMEKA	12.2
EEEKKK[K]EEEEKR	12.2
KLSEEE[K]KKAEEE	12.1
TLSEEE[K]EKLKKE	12.1

Fires in
46,327 / 64,998 designs

Energy (activation)
−13.5 kcal/mol q=0.011

Steering vs random
+2.8pp p<0.001 n=2184

Feature 6073

energy only

Leu preceded by acidic (D/E); 6 of top 12

Top activating sites	activation
KRAVQQ[L]RACGVL	73.5
QAFDEE[L]AFYGLV	36.1
SAYDEE[L]AFFGLI	35.5
PLYEEE[L]AFWGID	34.4
LDKLQV[E]RERGIT	21.7

Fires in
273 / 64,998 designs

Energy (presence)
−30.1 kcal/mol q=0.011

Steering vs random
+1.2pp p=1.00 n=84

Feature 11326

energy only

Basic (R/K) + Pro, before acidic/Pro-rich

Top activating sites	activation
KGDPGS[P]GPPGPA	18.4
EALRKA[R]ADPSVY	15.5
SADGKE[K]SPPAYL	15.1
TTAPDI[R]EDPKAF	15.1
RGEQGT[P]GPPGPA	14.2

Fires in
64,997 / 64,998 designs

Energy (activation)
−12.9 kcal/mol q=0.005

Steering vs random
+0.8pp p=0.21 n=1680